



Intercalation of small molecules in alkali metal fullerides superconductors

Renato C. da Silva^{1,2} · Cristiano C. Bastos³ · Antonio C. Pavão¹

Received: 17 January 2020 / Accepted: 20 March 2020
© Springer-Verlag GmbH Germany, part of Springer Nature 2020

Abstract

Using a BCS-adapted equation via multivariate analysis of structural and electronic properties, we calculated the critical temperature of fullerides A_2BC_{60} and A_3C_{60} , with pure A and B alkaline metal or solvated with NH_3 and CH_3NH_2 , in good agreement with the experimental data. In the ammoniation of Na_2CsC_{60} , concomitant expansion of the lattice and the increasing ionic character of the chemical bonds between the ligands and C_{60} , the critical temperature rises by about 20 K. For methylamine, there is an increase in critical temperature far lower than with ammonia, indicating that anisotropic lattice expansion does not favor the phenomenon. Besides crystal lattice expansion, a known factor that influences the critical temperature, our model identifies the other structural and electronic factors that contribute to the critical temperature of these materials. In alkali metal hydration, the crystal is expanded at the same time that the HOMO-LUMO gap is reduced, favoring electron transfer between the metal–ligand complex and C_{60} . With the intercalation of water molecules in these fullerides, we predict that Na_3C_{60} becomes superconducting at a critical temperature of 47.2 K and that Cs_3C_{60} reaches 50 K.

Keywords Fullerides superconductors · QSPR analysis · DFT calculations · Lattice expansion

1 Introduction

The explanation for superconductivity at high critical temperatures remains an intriguing theoretical–experimental challenge, and the discovery of superconducting fullerides has raised new questions about the phenomenon beyond those already posed involving superconducting metals, alloys, and oxides. With the synthesis of K_3C_{60} in 1991, the alkali metal fullerides drew attention both for being organic superconductors and for their high critical temperature (T_C), which reach values around 40 K for some alkali

metal compounds. Generally the structure of fullerides is face-centered cubic (fcc), with the molecules of C_{60} at the lattice points and the alkali metals occupying the tetrahedral and octahedral interstitial sites of the crystal. The tetrahedral site is approximately the size of the ion Na^+ , while the octahedral site is larger than any alkali ion. The lattice parameter value of alkali-doped fullerene is only about 1% higher than C_{60} , but such a small modification is crucial to favor the transition to the superconducting state. The expansion of the crystal with Na^+ is not enough to bring the fulleride to the superconducting state, but combinations with larger radius alkali ions have produced a variety of superconducting compounds. Structural spacers formed by the solvation of alkali ions with neutral molecules, such as ammonia [1] and methylamine [2], were also used for the purpose of expanding the crystal lattice and increasing the critical temperature. Zhou *et al.* modified crystal structure of Na_2CsC_{60} , with $T_C = 10.5$ K and lattice parameter $a = 14.132$ Å, solvating sodium ions with ammonia to produce the compound $(NH_3)_4Na_2CsC_{60}$ with $a = 14.473$ Å and $T_C = 29.6$ K [1]. NH_3 molecules form a tetrahedral structure with sodium ions, $Na(NH_3)_4^+$, such that the complex increases the effective radius of the octahedral sites of the crystal. However, the distance between C_{60} seems not to be the only

“Festschrift in honor of Prof. Fernando R. Ornellas” Guest Edited by Adélia Justino Aguiar Aquino, Antonio Gustavo Sampaio de Oliveira Filho and Francisco Bolivar Correto Machado.

✉ Antonio C. Pavão
pavao@ufpe.br

¹ Departamento de Química Fundamental, Universidade Federal de Pernambuco, Recife, PE 50740-540, Brazil

² Instituto Federal do Sertão Pernambucano, Ouricuri, PE 56200-000, Brazil

³ Departamento de Química, Universidade Federal Rural de Pernambuco, Recife, PE 52171-900, Brazil

relevant factor leading to the superconducting behavior of these fullerides. The significant increase in critical temperature, which almost tripled in the case of ammonia, indicates that in addition to the lattice expansion, modifications in the electronic structure introduced by the metal solvation acquire a major importance in the charge transport process. Potassium ammoniation forming the compound $(\text{NH}_3)\text{K}_3\text{C}_{60}$ also raises the critical temperature, though not so much, to 28 K [2]. However, the intercalation of molecules does not always lead to positive results. When potassium ions are mono-coordinated with CH_3NH_2 , the critical temperature of the $(\text{CH}_3\text{NH}_2)\text{K}_3\text{C}_{60}$ reduces to 11 K [2]. Therefore, besides the lattice parameter, there are other factors related to the structure and chemical behavior of the spacers that influence the critical temperature of these fullerides.

We have recently found that properties of constituent atoms play a relevant role in fulleride superconductivity [3]. In this work, besides the atomic properties, we analyze the influence of the electronic crystal properties. Using an extension of BCS theory that combines structural and electronic properties through the quantitative structure–property relationship (QSPR) methodology, we analyze various combinations of alkali metals solvated with small molecules to predict compounds with high T_C , some even reaching 50 K. The T_C was obtained using a multivariate analysis of the lattice parameter, volume, HOMO–LUMO gap, ionization energy of metals, and other electronic properties of crystals obtained from density functional theory (DFT) calculations. To describe variations in critical temperature, we analyzed both the increase in cell volume and the changes in the electronic structure with metal solvation. Our results are in good agreement with the experimental data for both alkali-doped fullerenes and the solvated compounds $(\text{NH}_3)_4\text{Na}_2\text{CsC}_{60}$, $(\text{CH}_3\text{NH}_2)\text{K}_3\text{C}_{60}$ and $(\text{NH}_3)\text{K}_3\text{C}_{60}$. We found that in the ammoniation of $\text{Na}_2\text{CsC}_{60}$, with the concomitant expansion of the lattice and the increase in the ionic character of the chemical bonds between metal–ligand complex and the C_{60} , the critical temperature rises by about 20 K. In the case of methylamine, there is an increase in critical temperature, but much lower than with ammonia, indicating that anisotropic lattice expansion does not favor the phenomenon. The results also show that the hydration of the Na_3C_{60} makes it superconducting, with a critical temperature of 47.2 K, and raises the temperature of the Cs_3C_{60} to 50 K.

2 Methodology

Since alkali-doped fullerenes behave as unconventional superconductors, it is necessary to explore extensions of the BCS theory beyond Migdal–Eliashberg to describe these compounds [4–6]. In this sense, different relations between the superconductors critical temperature and their structural

and electronic properties have been proposed [7–17], for example, relationships of T_C with the chemical composition [9–17], chemical potential [11], Debye temperature [11], boundary orbitals [12, 13], Mott–Hubbard energy [9, 14], ionic radius [9], chemical hardness [9], electron affinity [9], ionization energy [10, 15], electronegativity [18], electronic valence [8–10, 14–16], ionic character [9], atomic radius [8], Coulomb repulsion and conduction bandwidth [10, 14–16], and pressure [17, 19, 20]. To calculate the critical temperature of superconducting fullerides, we developed an extension of BCS theory through a QSPR multivariate analysis of structural and electronic properties such as lattice parameter, crystal HOMO–LUMO gap, crystal volume, electron affinity and ionization energy of the spacer molecules.

The BCS equation for calculating the T_C , $k_B T_C = B \hbar \omega_0 e^{\frac{N(0)V_0}{2}}$ presents two fundamental variables, the electron–phonon interaction (V_0) and the electron density at the Fermi level $N(0)$. k_B is the Boltzmann constant and $\hbar \omega_0$ is the energy of the phonon, which is approximately $k_B \Theta_M$, with Θ_M being the Debye temperature of the metal and B a coefficient to be determined. According to Pauling, it is possible to rewrite an approximation of this equation [8]

$$T_C = 120 e^{-\frac{1}{\rho}} \quad (1)$$

The T_C of pure and solvated fulleride crystals such as A_2BC_{60} or A_3C_{60} were calculated using equation 1, where ρ is obtained from the lattice parameter (a), HOMO–LUMO gap (Δ), cell volume (V), electron affinity, (A) and ionization energy (I) of the ligands. The selection of these properties was performed using a QSPR statistical analysis combined with multiple linear regression (MLR). From a representative training set of experimentally known fullerides, the QSPR methodology selected the variables for the calculation of $\rho = b_0 + b_1(X_1)^{\beta_1} + b_2(X_2)^{\beta_2} + \dots + b_k(X_k)^{\beta_k}$ where X_k are the descriptors of the structural and electronic properties. The exponents $\beta_1, \beta_2, \dots, \beta_k$ and the coefficients $b_0, b_1, b_2, \dots, b_k$ were adjusted through MLR, which is able to describe the correlation of two or more independent variables as a function of a dependent variable. In response to the correlation between the independent variables and the dependent variable, it is possible to predict the theoretical value of the dependent variable under study. The prediction is obtained from the mathematical model generated by MLR and the chosen adjustment for the relationships between the variables. In response to this prediction it is possible to identify which properties best describe the experimental properties under study. For the superconducting fullerides, the dependent variables used were ρ and the lattice parameter a of the crystal, both used for critical temperature calculation. MLR uses the Newton–Raphson numerical adjustment method, evaluated by the correlation coefficient (R), standard deviation (s), determination coefficient (R^2), and analysis

of the dispersion point in relation to the regression line in the graph. A significant QSPR statistical model, as was the case here, should have the highest possible value for the Fischer test F , within the minimum acceptable value found in the reference tables. A properly adjusted model should have R next to one and s close to zero. The coefficient of determination, which expresses the model's ability to explain variations in experimental values, must also be close to one for a suitably adjusted model.

2.1 Dft theory

The structural and electronic properties of fullerides were obtained by DFT calculations using the hybrid functional B3LYP that combines the exchange functional proposed by Becke with the Lee-Yang-Parr correlation functional. The B3LYP functional is defined as $B3LYP = 0.20Ex(HF) + 0.80Ex(Slater) + 0.72DEx(B88)^{GGA} + 0.19Ec(VWN)^{LDA} + 0.81DEc(LYP)^{GGA}$, where Ex is the exchange energy, Ec is the correlation energy, and HF, B88, VWN, and LYP are the Hartree-Fock, Beck, Vosko-Wilk-Nusair, and Lee-Yang-Parr terms, respectively. The term LYP represents the local and nonlocal correlations, while the excess of the local correlation is given by the expression VWN. The exchange functional B88 and the correlation functional LYP are the generalized gradient approximations (GGA), while VWN corresponds to the local density approximation (LDA). The functional hybrid is an attempt to overcome some difficulties of the pure functional. The exchange correlation term contain parameters that are adjusted to reproduce experimental data or *ab initio* results, such as asymptotic behavior over long distances. For alkali metals, the Stuttgart/Dresden potential SDD base was used, which proved to be suitable for these alkali atoms. For alkali metal-doped fullerene crystals, the LanL1mb base set of the Hay and Wadt potential [21] was used, which proved to be adequate to describe the systems studied here.

3 2.2 Cluster model

The structure of C_{60} corresponds to a truncated icosahedron, consisting of 12 pentagonal and 20 hexagonal faces. Its spheroidal geometric structure resembles a soccer ball and belongs to the highest point symmetry group I_h , with 120 symmetry operations in group theory. The crystal unit cell C_{60} is fcc with its 14 balls of 60 carbons, lattice parameter $a = 14.17 \text{ \AA}$, and symmetry $Fm3m$. The C_{60} doped with alkali metals has produced a superconducting material that still occupies a special place in solid state chemistry. The crystal fullerides A_2BC_{60} or A_3C_{60} also possess structure fcc and differ only by the lattice parameter. The smaller radius A metals lie precisely in the tetrahedral and octahedral sites. As far as B-metals, larger radii are located in the octahedral site. We designed different cluster models for the A_3C_{60} ($A = Na, K, Rb, Cs$) which are shown in Figure 1a: $Na_{21}(C_{60})_{14}$, $K_{21}(C_{60})_{14}$, $Rb_{21}(C_{60})_{14}$, $Cs_{21}(C_{60})_{14}$. Of the 21 atoms of A, eight are in the tetrahedral sites and thirteen in the octahedral sites. The clusters for the A_2BC_{60} ($A, B = Na, K, Rb, Cs$) can be seen in Figure 1b, as shown in: $Na_{14}K_7(C_{60})_{14}$, $Na_{14}Rb_7(C_{60})_{14}$, $Na_{14}Cs_7(C_{60})_{14}$, $K_{14}Rb_7(C_{60})_{14}$, $K_{14}Cs_7(C_{60})_{14}$, $Rb_{14}K_7(C_{60})_{14}$, $Rb_{14}Cs_7(C_{60})_{14}$, $Cs_{14}Rb_7(C_{60})_{14}$. Of the 14 atoms of A, eight are at the tetrahedral sites and six at the octahedral sites, while for B, the seven atoms are at octahedral sites.

The solvation of alkali metals with ammonia, meth-ylamine and water in fullerides A_2BC_{60} or A_3C_{60} was represented by cluster models $(NH_3)_4Na_{14}Cs_7(C_{60})_{14}$, $(NH_3)_3K_{21}(C_{60})_{14}$, $(CH_3NH_2)K_{21}(C_{60})_{14}$, where the solvated metal occupies the octahedral site of the crystal. The radius of the octahedral site in the structure fcc of crystal A_2BC_{60} or A_3C_{60} is given by $r_{oct} = 0.5a - r_{C60}$, where $r_{C60} = 3.55 \text{ \AA}$ [1]. Thus, according to our calculations, the volume of the solvated metal does not exceed this value. Figure 2a–d

Fig. 1 Cluster models: **a** A_3C_{60} ($A = Na, K, Rb, Cs$). Yellow and red indicate the tetrahedral and octahedral sites, respectively. Eight atoms of A are in the tetrahedral sites and thirteen in the octahedral sites; **b** A_2BC_{60} ($A, B = Na, K, Rb, Cs$), yellow and green indicate the tetrahedral and octahedral sites, respectively for metal A, and red indicate the octahedral sites for metal B. Eight atoms of A are at the tetrahedral sites and six at the octahedral sites, while for B, the seven atoms are at octahedral sites

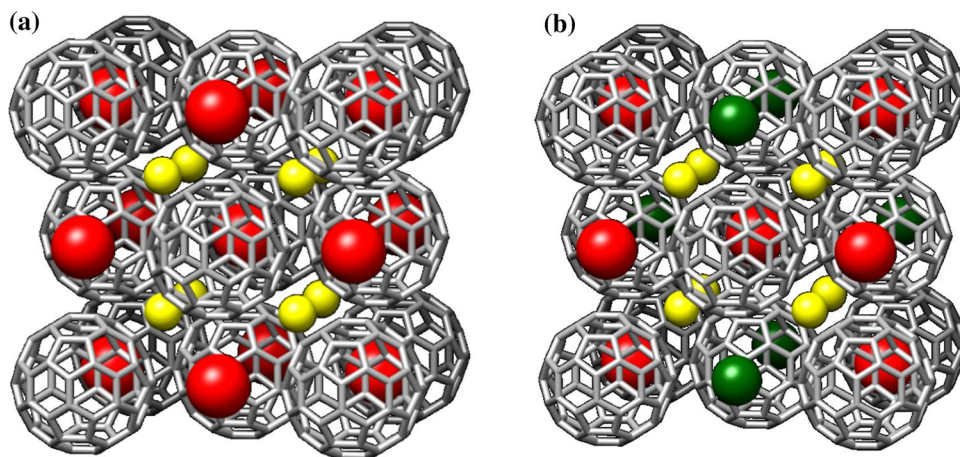
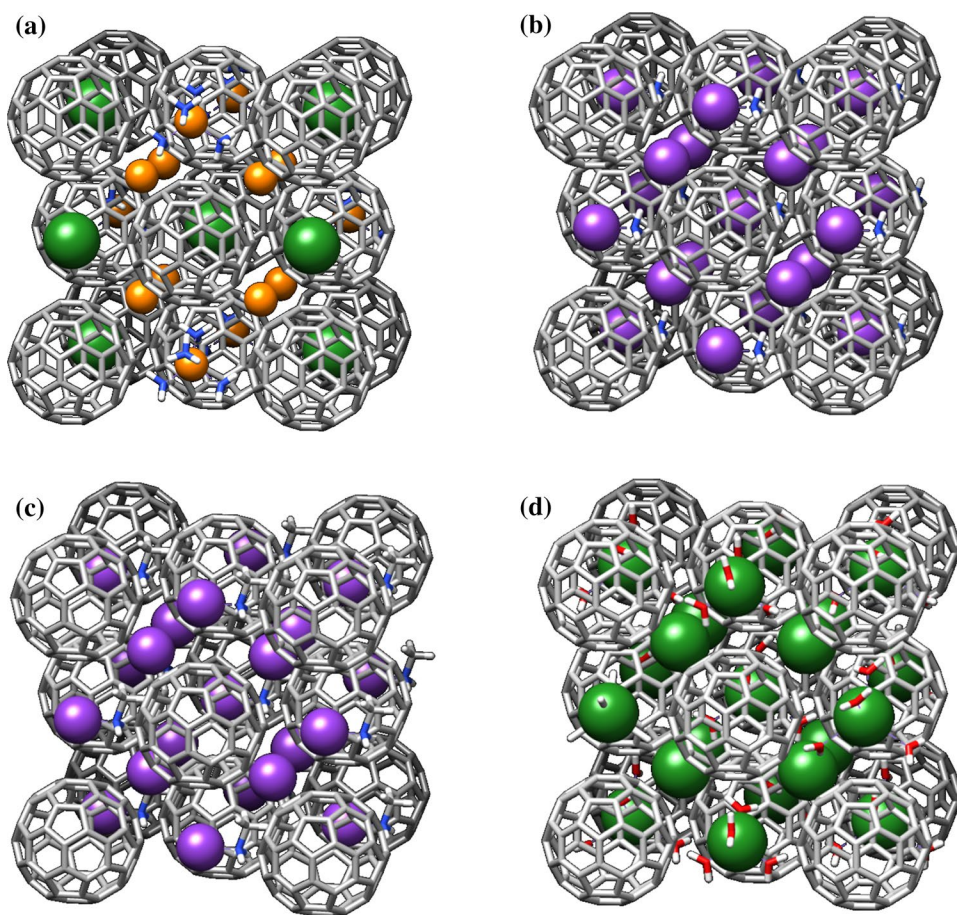


Fig. 2 Cluster model for solvated fullerides: **a** $(\text{NH}_3)_4\text{Na}_{14}\text{Cs}_7(\text{C}_{60})_{14}$; **b** $(\text{NH}_3)_4\text{K}_{21}(\text{C}_{60})_{14}$; **c** $(\text{CH}_3\text{NH}_2)_{14}\text{K}_{21}(\text{C}_{60})_{14}$; **d** $(\text{H}_2\text{O})_4\text{Cs}_{21}(\text{C}_{60})_{14}$ (Na)(K)(Cs). The solvated metal occupies the octahedral site of the cluster



shows some cluster models used in this work for pure and solvated fulleride crystals with neutral molecules.

4 Results and discussion

Table 1 presents the results of the DFT calculation for the HOMO-LUMO gap (Δ), the volume of the cluster (V), ionization energy (I), electron affinity (A) of the pure clusters (training set) and solvated (test set), the sum of ionization energy (I), and the electron affinity (A) of the isolated alkali metals.

Using the values from Table 1 in a QSPR analysis gives the following expression for the lattice parameter:

$$a = a_0 + a_1 \sum V + a_2 \left(\frac{\sum I + \sum A}{2} \right)^2 \quad (2)$$

where $a_0 = 10.5213$, $a_1 = 0.0041$, and $a_2 = 0.0221$. As expected, the lattice parameter depends directly on the crystal volume, but also reveals a relationship to M $\ddot{u}$ lliken's electronegativity of metals. Figure 3 shows the excellent agreement between the calculated lattice parameter and the experimental value for the training set.

The critical temperature was calculated using the lattice parameter (a), the HOMO-LUMO gap (Δ), the volume of the crystal (V) and the ionization energy (I) of the three alkali metals. The QSPR analysis led to the following expression:

$$\rho = b_0 - \frac{b_1}{a} \left(\frac{\Delta}{V} \right)^2 - b_2 \sum I \quad (3)$$

where $b_0 = 2.3705$, $b_1 = 1728269.3628$, and $b_2 = 0.1305$. Equation 3 describes a hypersurface of four variables, one with linear dependency and the other with quadratic and inverse dependencies, demonstrating a complex relationship between the selected properties and the critical temperature. The statistical results are quite reliable, with $R = 0.992$, $R^2 = 0.985$, $s = 0.03$, $F(2.8) = 257.27$. The validation test (R , $R^2 \sim 1$, $s < 0.05$, $F(2.8) > 4.46$) showed between 95–99% confidence. Cross-validation indicated that the model is highly predictable and statistically significant. The second term on the right side of equation 3 involves two structural variables, the lattice parameter and the crystal volume, combined with the HOMO-LUMO gap of the crystal. Dependence on variables a and V correctly shows that the T_C increases as the crystal lattice expands [1]. It is interesting to note that the quadratic dependence of Δ/V resembles

Table 1 HOMO-LUMO gap (Δ) and volume (V) of pure and solvated clusters, sum of the ionization energy (I), and electron affinity (A) of the alkali metals

Cluster	Δ Cluster (eV)	V Cluster ($\text{\AA}^3$)	ΣI Metal (eV)	ΣA Metal (eV)
<i>Training set</i>				
(C ₆₀) ₁₄	1.200	416.234	—	—
Na ₂₁ (C ₆₀) ₁₄	0.705	437.232	15.417	1.644
Na ₁₄ K ₇ (C ₆₀) ₁₄	0.647	495.530	14.619	1.597
Na ₁₄ Rb ₇ (C ₆₀) ₁₄	0.631	518.542	14.455	1.582
Na ₁₄ Cs ₇ (C ₆₀) ₁₄	0.614	550.759	14.172	1.567
K ₂₁ (C ₆₀) ₁₄	0.533	612.125	13.023	1.503
K ₁₄ Rb ₇ (C ₆₀) ₁₄	0.516	635.137	12.859	1.488
K ₁₄ Cs ₇ (C ₆₀) ₁₄	0.500	658.149	12.695	1.473
Rb ₁₄ K ₇ (C ₆₀) ₁₄	0.499	667.354	12.576	1.473
Rb ₂₁ (C ₆₀) ₁₄	0.482	681.161	12.531	1.458
Rb ₁₄ Cs ₇ (C ₆₀) ₁₄	0.466	713.379	12.248	1.443
Cs ₁₄ Rb ₇ (C ₆₀) ₁₄	0.450	745.596	11.965	1.428
Cs ₂₁ (C ₆₀) ₁₄	0.433	777.813	11.682	1.413
<i>Test set</i>				
(NH ₃) ₄ Na ₁₄ Cs ₇ (C ₆₀) ₁₄	0.581	801.066	12.040	1.900
(NH ₃)K ₂₁ (C ₆₀) ₁₄	0.501	710.512	12.347	1.940
(CH ₃ NH ₂)K ₂₁ (C ₆₀) ₁₄	0.995	730.102	12.994	1.966
(H ₂ O) ₄ Cs ₂₁ (C ₆₀) ₁₄	0.302	890.282	8.523	0.783
(H ₂ O) ₄ Na ₂₁ (C ₆₀) ₁₄	0.416	867.138	10.970	1.365

the harmonic oscillator potential, which has been used to describe electron–phonon coupling in metallic superconductors [8, 9]. According to the equation, T_C increases when the Δ decreases, which makes sense as the reduction favors the electron transfer. The latter term depends only on the electronic properties of metals, demonstrating their importance to the superconducting state. The linear dependence with ΣI contemplates the analysis of Hetfleisch *et al.*, which report a direct relationship between critical temperature and the sum of the ionization energies of the three alkali atoms in superconductors A₃C₆₀ and AB₂C₆₀ [10]. The inversely proportional dependence of the variable I with the T_C also appears in the present analysis; however, according to Eq. (1), it follows an exponential behavior, indicating that the analysis by Hetfleisch *et al.* corresponds to a limited region of the T_C curve versus I . Figure 4 shows the T_C compared to experimental data for the fulleride crystals already synthesized and the solvated compounds proposed here.

The calculated critical temperatures of (NH₃)₄Na₁₄Cs₇(C₆₀)₁₄ (31 K), (CH₃NH₂)K₂₁(C₆₀)₁₄ (13.5 K) and (NH₃)K₂₁(C₆₀)₁₄ (28.8 K) are consistent with the experimental values (29.6 K, 11 K e 28 K, respectively). It is important to highlight that these compounds were not included in the training set, indicating the reliability of the model to predict the T_C . Calculations with the clusters Na₁₄Cs₇(C₆₀)₁₄ ($a = 14.148 \text{ \AA}$) and (NH₃)₄Na₁₄Cs₇(C₆₀)₁₄, ($a = 14.879 \text{ \AA}$) show that the ammoniation not only does the lattice expand, but also narrows the HOMO-LUMO

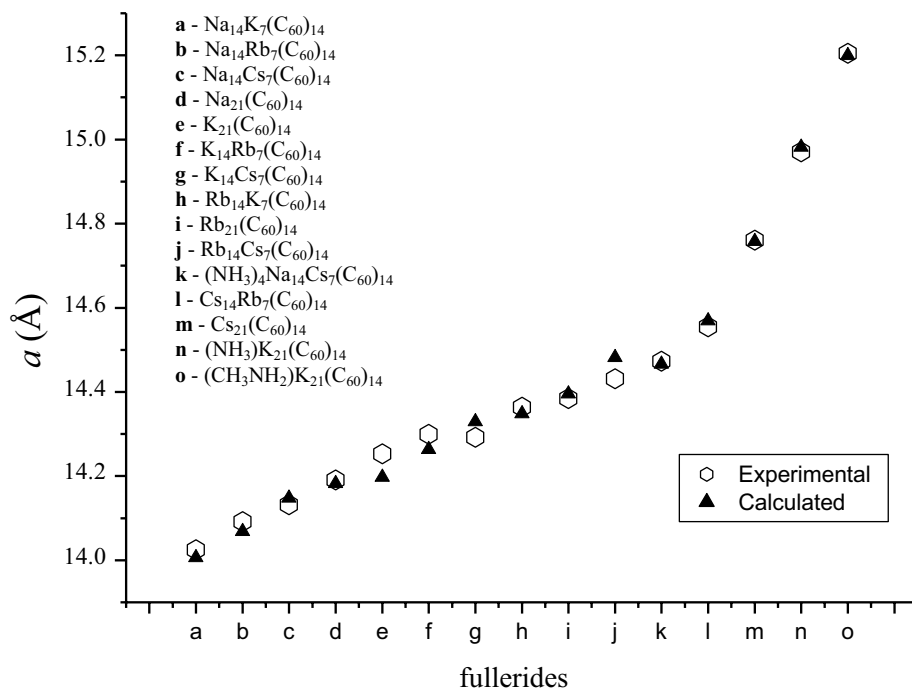
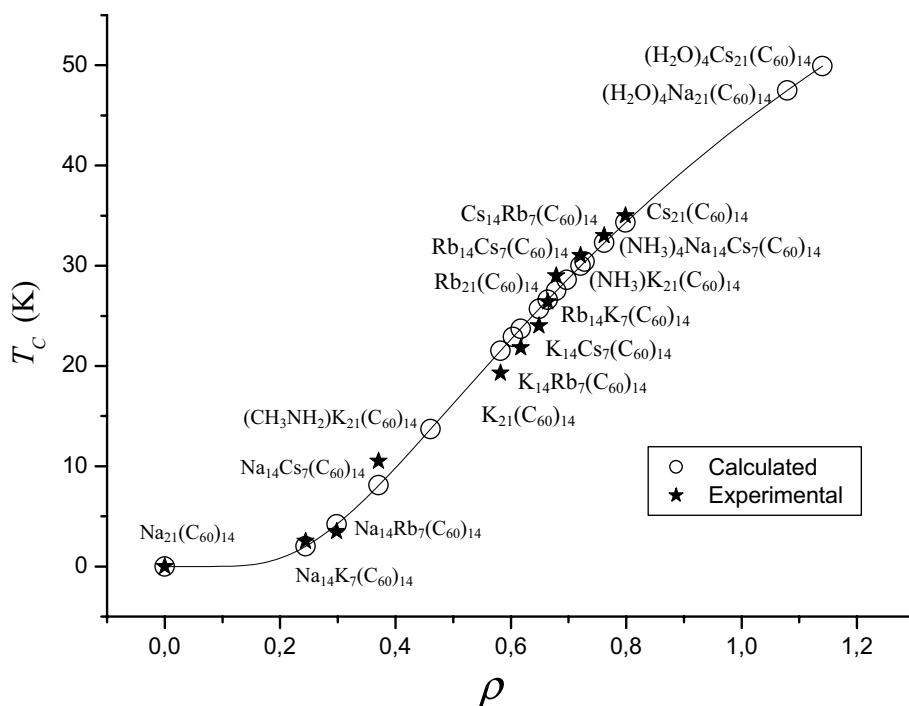
Fig. 3 Experimental and calculated lattice parameter for superconducting fullerides

Fig. 4 Experimental and calculated T_C for pure and solvated alkali metal fulleride superconductors



gap. The solvated cluster, $(\text{NH}_3)_4\text{Na}_{14}\text{Cs}_7(\text{C}_{60})_{14}$, has a higher volume ($801.01 \text{ \AA}^3$) and a smaller HOMO-LUMO gap (0.58 eV) compared to pure cluster, $\text{Na}_{14}\text{Cs}_7(\text{C}_{60})_{14}$, ($550.76 \text{ \AA}^3$ and 0.61 eV , respectively). These modifications in the structural and electronic properties are responsible for increasing the critical temperature around 20 K . For $(\text{NH}_3)\text{K}_{21}(\text{C}_{60})_{14}$, the HOMO-LUMO gap is 0.50 eV , and the volume is $710.51 \text{ \AA}^3$, so the lower volume and decrease of the HOMO-LUMO gap for the solvated cluster $(\text{NH}_3)_4\text{Na}_{14}\text{Cs}_7(\text{C}_{60})_{14}$ ($801.01 \text{ \AA}^3$ e 0.58 eV , respectively), leads to a decrease in the critical temperature of the $(\text{NH}_3)\text{K}_{21}(\text{C}_{60})_{14}$ (28.8 K). For $(\text{CH}_3\text{NH}_2)\text{K}_{21}(\text{C}_{60})_{14}$, although there is an increase in volume ($730.10 \text{ \AA}^3$) when compared to $(\text{NH}_3)\text{K}_{21}(\text{C}_{60})_{14}$ ($710.51 \text{ \AA}^3$), the critical temperature is reduced to 13.5 K . In this case, it is observed that the increase in HOMO-LUMO gap (0.50 to 0.99 eV) is responsible for decreasing the T_C , but anisotropic expansion in the crystal also has a significant effect [22]. Na_3C_{60} is not superconducting, and in fact, based on Eq. 1, its T_C is zero. However, the equation also shows that when sodium is coordinated with NH_3 , the critical temperature is 21.8 K . When sodium is combined with H_2O , forming a tetra-coordinate complex, the critical temperature of the $(\text{H}_2\text{O})_4\text{Na}_{21}(\text{C}_{60})_{14}$ rises to 47.2 K . When H_2O is combined with Cs^+ at $(\text{H}_2\text{O})_4\text{Cs}_{21}(\text{C}_{60})_{14}$, the critical temperature rises to 49.8 K . For all these compounds there is lattice expansion, increase in electronegativity difference and decrease of HOMO-LUMO gap, which explains the critical high temperatures of these alkali metal fullerides.

5 Conclusions

Using the adapted BCS equation and multivariate analysis of structural and electronic properties (ionization energy and electron affinity of alkali metals, the HOMO-LUMO and volume of different cluster models), it was possible to calculate the critical temperature of fullerides A_2BC_{60} and A_3C_{60} , with pure metal or solvated with NH_3 and CH_3NH_2 , in good agreement with experimental data. With the intercalation of water molecules in these fullerides, we increase the critical temperature to about 50 K . Beyond crystal lattice expansion, a known factor that influences the T_C , our model identifies the other electronic factors that contribute to the critical temperature of these materials. We found that with proper hydration of the alkali metal, the crystal is expanded at the same time as the HOMO-LUMO gap reduces and favors electron transfer between the metal–ligand complex and the C_{60} . The synthesis of the compounds suggested here represents an experimental challenge, but the expressive results indicated stimulate the search for superconductors with higher critical temperatures. As a more general conclusion, we can say that by identifying properties connected with the superconductivity of fullerides, our model offers a contribution to the understanding of superconductors as a whole and opens the way for future work and perspectives involving superconducting metals, alloys and oxides.

Acknowledgements The authors acknowledge the CAPES and the CENAPAD-SP.

References

- Zhou O, Fleming RM, Murphy DW, Rosseinsky MJ, Ramirez AP, van Dover RB, Haddon RC (1993) Increased transition temperature in superconducting $\text{Na}_2\text{CsC}_{60}$ by intercalation of ammonia. *Nature* 362:433–435. <https://doi.org/10.1038/362433a0>
- Takabayashi Y, Prassides K (2016) Unconventional high- T_c superconductivity in fullerides. *Philos Trans R Soc A* 374:20150320. <https://doi.org/10.1098/rsta.2015.0320>
- Silva RC, Bastos CC, Pavao AC (2019) High- T_c fullerides. *Physica C: Supercond Appl*. 561:13–17. <https://doi.org/10.1016/j.physc.2019.02.001>
- Eliashberg GM (1960) Interactions between electrons and lattice vibrations in a superconductor. *Sov Phys JETP* 11:696–702
- McMillan WL (1968) Transition temperature of strong-coupled superconductors. *Phys Rev* 167:331–344. <https://doi.org/10.1103/PhysRev.167.331>
- Akashi R, Arita R (2013) Nonempirical study of superconductivity in alkali-doped fullerides based on density functional theory for superconductors. *Phys Rev B* 88:054510. <https://doi.org/10.1103/PhysRevB.88.054510>
- Bardeen J, Cooper LN, Schrieffer JR (1957) Theory of Superconductivity. *Phys Rev* 108:1175–1204. <https://doi.org/10.1103/PhysRev.108.1175>
- Pauling L (1968) The Resonating-valence-bond theory of superconductivity: crest superconductors and trough superconductors. *Proc Natl Acad Sci USA* 60:59–65. <https://doi.org/10.1073/pnas.60.1.59>
- Pauling L (1991) The structure of K_3C_{60} and the mechanism of superconductivity. *Proc Natl Acad Sci USA* 88:9208–9209. <https://doi.org/10.1073/pnas.88.20.9208>
- Hetfleisch F, Stepper M, Roeser HP, Bohr A, Lopez JS, Mashmool M, Roth S (2015) A correlation between ionization energies and critical temperatures in Superconducting A_3C_{60} fullerides. *Physica C* 513:1–3. <https://doi.org/10.1016/j.physc.2015.02.048>
- Webb GW, Marsiglio F, Hirsch JE (2015) Superconductivity in the elements. *Alloys Simple Compd Phys C* 514:17–27. <https://doi.org/10.1016/j.physc.2015.02.037>
- Tanaka J (2006) Ab initio quantum chemical calculation of the pair potentials of superconductors. *Physica C* 445–448:150–153. <https://doi.org/10.1016/j.physc.2006.06.039>
- Makino Y, Yoshimura K (2014) Empirical understanding of superconducting critical temperature based on valence electron parameters. *Physica C* 499:24–35. <https://doi.org/10.1016/j.physc.2014.01.006>
- Gunnarsson O (1997) Superconductivity in fullerides. *Rev Mod Phys* 69:575–606. <https://doi.org/10.1103/RevModPhys.69.575>
- Ramirez AP (2015) Superconductivity in alkali-doped C_{60} . *Physica C* 514:166–172. <https://doi.org/10.1016/j.physc.2015.02.014>
- Larsson S (2014) T_c Dependence on Hubbard U in the fullerenes. *J Supercond Novel Mag* 28:315–317. <https://doi.org/10.1007/s10948-014-2881-9>
- Hetfleisch F, Gunnarsson O, Srama R, Han JE, Stepper M, Roeser H-P, Bohr A, Lopez JS, Mashmool M, Roth S (2018) Chemical effects of alkali atoms on critical temperature in superconducting alkali-doped fullerides. *Physica C* 546:34–43. <https://doi.org/10.1016/j.physc.2017.12.005>
- Buzea C, Yamashita T (2000) Correlation between electronegativity and superconductivity. *Physica B* 281–282:951–952. [https://doi.org/10.1016/S0921-4526\(99\)01202-8](https://doi.org/10.1016/S0921-4526(99)01202-8)
- Diederichs J, Schilling JS, Herwig KW, Yelon WB (1997) Dependence of the superconducting transition temperature and lattice parameter on hydrostatic pressure for Rb_3C_{60} . *J Phys Chem Solids* 58:123–132. [https://doi.org/10.1016/S0022-3697\(96\)00087-X](https://doi.org/10.1016/S0022-3697(96)00087-X)
- Margadonna S, Brown CM, Lappas A, Prassides K, Tanigaki K, Knudsen KD, Bihan TL, Mézouar M (1999) Pressure and temperature evolution of the structure of the superconducting $\text{Na}_2\text{CsC}_{60}$ fulleride. *J Solid State Chem* 145:471–478. <https://doi.org/10.1006/jssc.1998.8159>
- Frisch MJ, Trucks GW, Schlegel HB, Scuseria GE, Robb MA, Cheeseman JR, Scalmani G, Barone V, Mennucci B, Petersson GA, Nakatsuji H, Caricato M, Li X, Hratchian HP, Izmaylov AF, Bloino J, Zheng G, Sonnenberg JL, Hada M, Ehara M, Toyota K, Fukuda R, Hasegawa J, Ishida M, Nakajima T, Honda Y, Kitao O, Nakai H, Vreven T, Montgomery JA Jr, Peralta JE, Ogliaro F, Bearpark M, Heyd JJ, Brothers E, Kudin KN, Staroverov VN, Kobayashi R, Normand J, Raghavachari K, Rendell A, Burant JC, Iyengar SS, Tomasi J, Cossi M, Rega N, Millam JM, Klene M, Knox JE, Cross JB, Bakken V, Adamo C, Jaramillo J, Gomperts R, Stratmann RE, Yazyev O, Austin AJ, Cammi R, Pomelli C, Ochterski JW, Martin RL, Morokuma K, Zakrzewski VG, Voth GA, Salvador P, Dannenberg JJ, Dapprich S, Daniels AD, Farkas Ö, Foresman JB, Ortiz JV, Cioslowski J, Fox DJ (2009) Gaussian 09, revision D.01, Gaussian Inc. (2009) Wallingford CT
- Arcon D, Pregelj M, Zorko A, Ganin AY, Rosseinsky MJ, Takabayashi Y, Prassides K (2008) Antiferromagnetic resonance in methylaminated potassium fullerides $(\text{CH}_3\text{NH}_2)_\text{x}\text{K}_3\text{C}_{60}$. *Phys Rev B* 77:35104–35106. <https://doi.org/10.1103/PhysRevB.77.035104>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.